

11(a)	generates <u>ultrasound</u>	B1
	detects <u>reflected</u> ultrasound	B1
	applied p.d. causes crystal to vibrate or vibrations cause crystal to generate an e.m.f.	B1
11(b)(i)	product of density and speed	M1
	speed of ultrasound in medium	A1
11(b)(ii)	difference between (the specific acoustic impedances)	C1
	• if similar/same then reflection coefficient is zero/very low • if very different then reflection coefficient is (nearly) 1 • the lower the difference means lower the reflection coefficient (<i>any one point</i>)	A1